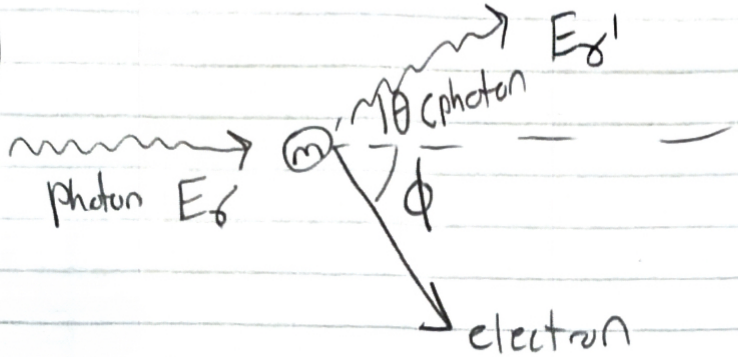
E&M Homework #11

Parker Lewis

Prob 1.)



- $p^\alpha = \left(\frac{E}{c}, \vec{p}\right)$

$E = \sqrt{c^2p^2 + m^2c^4}$ · $p^\alpha = (\gamma mc, \gamma m\vec{u})$

$E^2 = c^2p^2 + m^2c^4$

photon:

$m_\gamma \approx 0$

- $\hat{p}_\gamma \cdot \hat{p}_\gamma' = \cos\theta$ $\quad E^2 = c^2p^2$
- $\hat{p}_\gamma \cdot \hat{p}_e' = \cos\phi$ $\quad \vec{p}_\gamma = \frac{E_\gamma}{c}\hat{p}_\gamma$ $\quad p = \frac{E}{c}$

$\vec{p}_\gamma' = \frac{E_\gamma'}{c}\hat{p}_\gamma'$

$p^\gamma = \left(\frac{E_\gamma}{c}, \frac{E_\gamma}{c}\hat{p}_\gamma\right)$

$p^e = \left(\frac{mc^2}{c}, \vec{0}\right)$

$p^{\gamma'} = \left(\frac{E_\gamma'}{c}, \frac{E_\gamma'}{c}\hat{p}_\gamma'\right)$

$p^{e'} = \left(\frac{E_e'}{c}, \vec{p}_e'\right) = \left(\frac{E_e'}{c}, \sqrt{2mT}\,\hat{p}_e'\right)$

) $B \cdot A = B_\alpha A^\alpha = B^\circ A^\circ - \vec{B} \cdot \vec{A}$

$$\left[\left(\frac{E_\gamma}{c}, \frac{E_\gamma}{c}\hat{P}_\gamma\right) - \left(\frac{E_e'}{c}, \sqrt{2mT}\,\hat{P}_e'\right)\right]^2$$

$$= \left[\left(\frac{E_\gamma'}{c}, \frac{E_\gamma'}{c}\hat{P}_{\gamma'}\right) - \left(mc, \vec{0}\right)\right]^2$$

$$\left(\frac{E_\gamma}{c}, \frac{E_\gamma}{c}\hat{P}_\gamma\right)^2 - 2\left(\frac{E_\gamma}{c}, \frac{E_\gamma}{c}\hat{P}_\gamma\right)\left(\frac{E_e'}{c}, \sqrt{2mT}\,\hat{P}_e'\right) + \left(\frac{E_e'}{c}, \sqrt{2mT}\,\hat{P}_e'\right)^2$$

$$\left(\frac{E_\gamma}{c}, \frac{E_\gamma}{c}\hat{P}_\gamma\right)^2$$

$$= \frac{E_\gamma^2}{c^2} - \frac{E_\gamma^2}{c^2} = 0$$

$$\left(\frac{E_e'}{c}, \sqrt{2mT}\,\hat{P}_e'\right)^2$$

$$= \frac{E_e'^2}{c^2} - P_e'^2 = \frac{E_e'^2}{c^2} - \left(\frac{E_e'^2}{c^2} - m^2c^2\right)$$

$$= m^2c^2$$

$$-2\left(\frac{E_\gamma}{c}, \frac{E_\gamma}{c}\hat{P}_\gamma\right)\left(\frac{E_e'}{c}, \sqrt{2mT}\cdot\hat{P}_e'\right)$$

$$= -2\left[\frac{E_\gamma E_e'}{c^2} - \frac{E_\gamma}{c}\sqrt{2mT}\cos\phi\right]$$

$$= \frac{-2E_\gamma E_e'}{c^2} + \frac{2E_\gamma}{c}\sqrt{2mT}\cos\phi$$

$$\frac{E_e'^2}{c^2} + m^2c^2 - \frac{2E_\gamma E_e'}{c^2} + \frac{2E_\gamma}{c}\sqrt{2mT}\cos\phi$$

$$= \left(\frac{E_\gamma'}{c}, \frac{E_\gamma'}{c}\hat{P}_\gamma'\right)^2 - 2\left(\frac{E_\gamma'}{c}, \frac{E_\gamma'}{c}\hat{P}_\gamma'\right)\left(mc, \vec{0}\right)$$

$$+ \left(mc, \vec{0}\right)^2$$

$$= \frac{E_\gamma'^2}{c^2} - \frac{E_\gamma^2}{c^2} - 2\left[\frac{E_\gamma' mc}{c}\right] + m^2c^2$$

$$\frac{E_\gamma}{c}\sqrt{2mT}\cos\phi = \frac{E_\gamma E_e'}{c^2} - mE_\gamma'$$

$$2mT = \left(\frac{c}{\cos\phi E_\gamma} \left(\frac{E_\gamma E_e'}{c^2} - m E_{\gamma'} \right) \right)^2$$

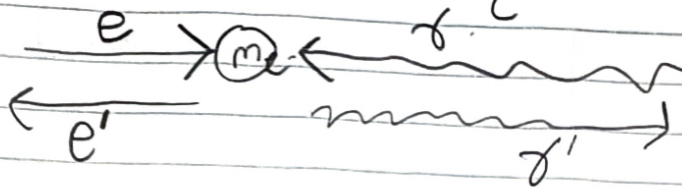
$$T = \frac{c^2}{2m E_\gamma^2 \cos^2\phi} \left(\frac{E_\gamma E_e'}{c^2} - m E_{\gamma'} \right)^2$$

• $\hat{P}_o = \hat{P}_o'$

• $\left(\frac{E}{c}, \vec{P}\right)$

problem 2.)

• $\vec{P}_o' = \frac{E_o'}{c}\hat{P}_o'$, $\vec{P}_o = \frac{E_o}{c}\hat{P}_o'$



• $\hat{P}_e \cdot \hat{P}_o' = \cos\theta$

• Total elastic collision • $E_e' \approx mc^2$ • $\vec{P}_e' \approx \vec{P}_e$

• θ: Angle between incident electron beam and scattered photon.

• $p^e \rightarrow$ incident electron 4-momentum.

• $p^{e'} \rightarrow$ scattered electron 4-mom.

• $p^\gamma \rightarrow$ in photon 4-mom.

• $p^{\gamma'} \rightarrow$ scattered Photon 4-mom.

• $p^e = \left(\frac{E_e}{c}, \vec{P}_e\right)$

• $p^{e'} = \left(\frac{E_e'}{c}, \vec{P}_e'\right) = \left(mc, \vec{P}_e\right)$

• $p^\gamma = \left(\frac{E_o}{c}, \vec{P}_o\right) = \left(\frac{E_o}{c}, \frac{E_o}{c}\hat{P}_o'\right)$

• $p^{\gamma'} = \left(\frac{E_o'}{c}, \vec{P}_o'\right) = \left(\frac{E_o'}{c}, \frac{E_o'}{c}\hat{P}_o'\right)$

$$\left(\frac{E_e}{c}, \vec{P_e}\right) + \left(\frac{E_0}{c}, -\frac{E_0}{c}\hat{P_0'}\right)$$

$$= \left(mc, -\vec{P_e}\right) + \left(\frac{E_0'}{c}, \frac{E_0'}{c}\hat{P_0'}\right)$$

$$\frac{E_e^2}{c^2} - \cancel{P_e^2} + 2\left(\frac{E_e E_0}{c^2} + \frac{E_0}{c} P_e \cos\theta\right)$$

$$= m^2c^2 - \cancel{P_e^2} + 2\left(\frac{mcE_0'}{c} + \frac{E_0'}{c} P_e \cos\theta\right)$$

$$\frac{E_e^2}{c^2} + \frac{2E_eE_0}{c^2} + \frac{2E_0 P_e \cos\theta}{c} = m^2c^2 + 2mE_0' + \frac{2E_0' P_e \cos\theta}{c}$$

$$E_0' = \frac{c}{2(mc + P_e\cos\theta)}\left[\frac{1}{c^2}\left(E_e^2 + 2\left(E_0E_e + \frac{E_0^2 P_e \cos\theta}{c}\right)\right) - m^2c^2\right]$$

$$E_0 = 1\,eV = 1.6\times10^{-19}\,J$$
$$E_e = 1\,GeV = 1\times10^9\,eV = 1.602\times10^{-10}\,J$$
$$m_e = 9.11\times10^{-31}\,kg$$
$$P_e = 5.34\times10^{-19}\ kg\cdot m/s$$

$$E_0' = -2.93\times10^{11}\,J = -1.83\times10^{30}\,eV = -1.83\times10^{21}\,GeV$$

Small approx $\cos\theta \approx 1 + \frac{\theta^2}{2}$

$E \approx -7.45 \times 10^{10}$ J

$\cos\theta \approx 1 - \frac{\pi^2}{2} \approx -3.934$

(Small approx)

$E = -4.65 \times 10^{29}$ eV $= -4.65 \times 10^{20}$ GeV

(Small

E&M HwTH 11

Parker Lewis

prob 3.)

a.) Prove $\vec{E}\cdot\vec{B}$ is Lorentz invariant.

Jackson 11.148:

- $E_1' = E_1$, $B_1' = B_1$
- $E_2' = \gamma(E_2 - \beta B_3)$, $B_2' = \gamma(B_2 + \beta E_3)$
- $E_3' = \gamma(E_3 + \beta B_2)$, $B_3' = \gamma(B_3 - \beta E_2)$

$$\vec{E}'\cdot\vec{B}' = E_1'B_1' + E_2'B_2' + E_3'B_3'$$

$$= E_1B_1 + \gamma^2(E_2 - \beta B_3)(B_2 + \beta E_3) + \gamma^2(E_3 + \beta B_2)(B_3 - \beta E_2)$$

$$= E_1B_1 + \gamma^2(E_2B_2 + \beta E_2E_3 - \beta B_3B_2 - \beta^2 E_3B_3) + \gamma^2(E_3B_3 - \beta E_2E_3 + \beta B_2B_3 - \beta^2 B_2E_2)$$

$$= E_1B_1 + \gamma^2E_2B_2 + \cancel{\gamma^2\beta E_2E_3} - \cancel{\gamma^2\beta B_3B_2} - \gamma^2\beta^2E_3B_3 + \gamma^2E_3B_3 - \cancel{\beta\gamma^2E_2E_3} + \cancel{\gamma^2\beta B_2B_3} - \gamma^2\beta^2B_2E_2$$

$$= E_1B_1 + \gamma^2(E_2B_2 + E_3B_3 - \beta^2(E_3B_3 + B_2E_2))$$

$$\gamma^2(1-\beta^2) = \frac{1}{(1-\beta^2)}(1-\beta^2) = 1$$

$$= E_1B_1 + (E_2B_2 + E_3B_3)\,\gamma^2(1-\beta^2)$$

$$= E_1B_1 + E_2B_2 + E_3B_3 = \vec{E}\cdot\vec{B}$$

$$\vec{E}\,\vec{B} = \vec{E}_t \cdot \vec{B}$$

• Electric and magnetic field and magnetic field correlate with one another

$(\vec{E}\times\vec{B})=0$, $(\vec{B}\times\vec{E})=0$

b.) $\vec{B}' = \gamma\left(\vec{B} - \frac{1}{c}(\vec{\beta}\times\vec{E})\right) - \frac{\gamma^2}{\gamma+1}\vec{\beta}\left(\vec{\beta}\cdot\vec{B}\right)$

$|\vec{B}'|^2 = B'^2$

$$B'^2 = \gamma^2\left(\vec{B} - \frac{1}{c}(\vec{\beta}\times\vec{E})\right)^2 + \frac{\gamma^4}{(\gamma+1)^2}\left(\vec{\beta}(\vec{\beta}\cdot\vec{B})\right.$$

$$- \frac{2\gamma^3}{(\gamma+1)}\left(\vec{B} - \frac{1}{c}(\vec{\beta}\times\vec{E})\right)\cdot\vec{\beta}\left(\vec{\beta}\cdot\vec{B}\right)$$

$$\left(\vec{B} - \frac{1}{c}(\vec{\beta}\times\vec{E})\right)\cdot\left(\vec{B} - \frac{1}{c}(\vec{\beta}\times\vec{E})\right)$$

$$= \vec{B}\cdot\vec{B} - \vec{B}\cdot\frac{1}{c}(\vec{\beta}\times\vec{E}) - \frac{1}{c}(\vec{\beta}\times\vec{E})\cdot\vec{B}$$

$$+ \frac{1}{c^2}(\vec{\beta}\times\vec{E})\cdot(\vec{\beta}\times\vec{E})$$

$$= B^2 - \frac{1}{c}(\vec{E}\times\vec{B})\cdot\vec{\beta} - \frac{1}{c}(\vec{E}\times\vec{B})\cdot\vec{\beta}$$

$$+ \frac{1}{c^2}\left[(\vec{\beta}\cdot\vec{\beta})(\vec{E}\cdot\vec{E}) - (\vec{\beta}\cdot\vec{E})(\vec{\beta}\cdot\vec{E})\right]$$

$$= B^2 + \frac{1}{c^2}\left(\beta^2E^2 - (\vec{\beta}\cdot\vec{E})^2\right)$$

$$\cdot \frac{\gamma^4}{(\gamma+1)^2}\left(\vec{\beta}(\vec{\beta}\cdot\vec{B})\right)^2$$

$$= \frac{\gamma^4\beta^2(\vec{\beta}\cdot\vec{B})^2}{(\gamma+1)^2}$$

$$\cdot \frac{2\gamma^3}{(\gamma+1)}\left(\vec{B} - \frac{1}{c}(\vec{\beta}\times\vec{E})\right)\cdot\vec{\beta}(\vec{\beta}\cdot\vec{B})$$

$$= \frac{2\gamma^3}{(\gamma+1)}\left[\vec{B}\cdot\vec{\beta}(\vec{\beta}\cdot\vec{B}) - \frac{1}{c}\vec{\beta}(\vec{\beta}\cdot\vec{B})\cdot(\vec{\beta}\times\vec{E})\right]$$

$$= \frac{2\gamma^3}{(\gamma+1)}\left[\vec{B}\cdot\vec{\beta}(\vec{\beta}\cdot\vec{B}) - \frac{\vec{\beta}}{c}\left(\cancel{\vec{\beta}(\vec{\beta}\cdot\vec{B})\times\vec{E}}\right)\right]$$

$$= \frac{2\gamma^3}{(\gamma+1)}\vec{B}\cdot\vec{\beta}(\vec{\beta}\cdot\vec{B})$$

$$B'^2 = \gamma^2\vec{B}^2 + \frac{\gamma^2}{c^2}\left(\beta^2E^2 - (\vec{\beta}\cdot\vec{E})^2\right) - \frac{2\gamma^3}{(\gamma+1)}\vec{\beta}\cdot\vec{\beta}(\vec{\beta}\cdot\vec{B})$$

$$+ \frac{\gamma^4\beta^2(\vec{\beta}\cdot\vec{B})^2}{(\gamma+1)^2}$$

$$\vec{E}'^2 = \gamma^2\left(\vec{E} + c\vec{\beta}\times\vec{B}\right)^2 + \frac{\gamma^4}{(\gamma+1)^2}\beta^2(\vec{\beta}\cdot\vec{E})^2$$

$$- \frac{2\gamma^3}{(\gamma+1)}\left(\vec{E} + c(\vec{\beta}\times\vec{B})\right)\cdot\vec{\beta}(\vec{\beta}\cdot\vec{E})$$

$$\gamma^2\left(\vec{E} + c\vec{\beta}\times\vec{B}\right)^2$$

$$\gamma^2\left(\vec{E} + c\vec{\beta}\times\vec{B}\right)\cdot\left(\vec{E} + c\vec{\beta}\times\vec{B}\right)$$

$$= \gamma^2\left[E^2 + \vec{E}\cdot(c\vec{\beta}\times\vec{B}) + (c\vec{\beta}\times\vec{B})\cdot\vec{E} + (c\vec{\beta}\times\vec{B})\cdot(c\vec{\beta}\times\vec{B})\right]$$

$$= \gamma^2\left[E^2 + (c\vec{\beta}\cdot c\vec{\beta})(\vec{B}\cdot\vec{B}) - (c\vec{\beta}\cdot\vec{B})(c\vec{\beta}\cdot\vec{B})\right]$$

$$= \gamma^2\left[E^2 + c^2\beta^2B^2 - (c\vec{\beta}\cdot\vec{B})^2\right]$$

$$\frac{2\gamma^3}{(\gamma+1)}\vec{\beta}(\vec{\beta}\cdot\vec{E})\cdot\left(\vec{E}+(\vec{\beta}\times\vec{B})\right)$$

$$\frac{2\gamma^3}{(\gamma+1)}\left[\vec{\beta}(\vec{\beta}\cdot\vec{E})\cdot\vec{E} + \cancel{\vec{\beta}(\vec{\beta}\cdot\vec{E})\cdot(\vec{\beta}\times\vec{B})}\right]$$

$$= \frac{2\gamma^3}{(\gamma+1)}\vec{\beta}(\vec{\beta}\cdot\vec{E})$$

$$E'^2 = \gamma^2E^2 + \gamma^2c^2\left(\beta^2B^2 - (\vec{\beta}\cdot\vec{B})^2\right) - \frac{2\gamma^3}{(\gamma+1)}\vec{\beta}(\vec{\beta}\cdot\vec{E})\cdot\vec{E} + \frac{\gamma^4}{(\gamma+1)}\beta^2(\vec{\beta}\cdot\vec{E})^2$$

$|\vec{B}| = \frac{|\vec{E}|}{c}$

$$\frac{E'^2}{c^2} = \frac{\gamma^2}{c^2}E^2 + \gamma^2\left(\beta^2B^2 - (\vec{\beta}\cdot\vec{B})^2\right) - \frac{2\gamma^3}{(\gamma+1)}\vec{\beta}\frac{(\vec{\beta}\cdot\vec{E})}{c^2} + \frac{\gamma^4}{(\gamma+1)}\beta^2\frac{(\vec{\beta}\cdot\vec{E})^2}{c^2}$$

Subtract

$$\therefore \frac{E'^2}{c^2} - B'^2 = \frac{\gamma^2}{c^2}E^2 + \gamma^2\left(\beta^2B^2 - \cancel{(\vec{\beta}\cdot\vec{B})^2}\right) - \frac{2\gamma^3}{(\gamma+1)}\frac{\vec{\beta}(\vec{\beta}\cdot\vec{E})}{c^2}\cdot\vec{E} + \cancel{\frac{\gamma^4\beta^2(\vec{\beta}\cdot\vec{E})^2}{(\gamma+1)^2c^2}} - \gamma^2B^2 - \frac{\gamma^2}{c^2}\left(\beta^2E^2 - \cancel{(\vec{\beta}\cdot\vec{E})^2}\right) + \cancel{\frac{2\gamma^3}{(\gamma+1)}\vec{\beta}\cdot\vec{\beta}(\vec{\beta}\cdot\vec{B})} - \cancel{\frac{\gamma^4\beta^2(\vec{\beta}\cdot\vec{B})^2}{(\gamma+1)^2}}$$

• in vaccum $|B| = \frac{|E|}{c}$

• $\frac{E'^2}{c^2} - B'^2 = \frac{\gamma^2}{c^2}E^2 + \gamma^2\beta^2B^2 - \gamma^2B^2 - \frac{\gamma^2}{c^2}\beta^2E^2$

• $\frac{E'^2}{c^2} - B'^2 = \gamma^2\left(\frac{E^2}{c^2} - \frac{E^2\beta^2}{c^2}\right) - \gamma^2\left(B^2 - \beta^2B^2\right)$

$= \cancel{\gamma^2\left(1-\beta^2\right)\frac{E^2}{c^2}} - \cancel{\gamma^2\left(1-\beta^2\right)B^2}$

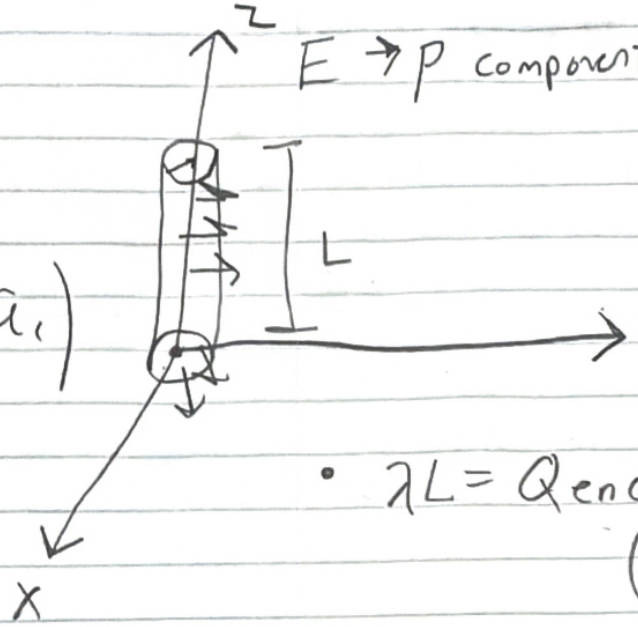
• $\frac{E'^2}{c^2} - B'^2 = \frac{E^2}{c^2} - B^2$

Prob 4.)

- Consider an infinite line charge λ on z-axis.

(ct, x, y, z) (line charge is rest in this frame.

a.)



- $E \rightarrow p$ component
- $K \rightarrow$ Rest frame
- $K' \rightarrow$ moving frame

Cylindrical coordinates:

- (ρ, ϕ, z)
- $\lambda L = Q_{enc}$ (In Rest frame)

- $\int_A \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}$
- $0 \leq z \leq L$
- $0 \leq \phi \leq 2\pi$
- $d\vec{A} = \rho \, d\phi \, dz \, \hat{\rho}$

$$\rho E_\rho \int_0^{2\pi} \int_0^L d\phi \, dz = \frac{\lambda L}{\varepsilon_0}$$

$$\rho E_\rho L 2\pi = \frac{\lambda L}{\varepsilon_0}$$

$$E_\rho = \frac{\lambda}{2\pi\rho\varepsilon_0}$$

$$\boxed{\vec{E} = \frac{\lambda}{2\pi\rho\varepsilon_0}\hat{\rho}}$$

$$\vec{B} = \gamma\left(\vec{B} - \frac{1}{c}\vec{\beta}\times\vec{E}\right) - \frac{\gamma^2}{\gamma+1}\vec{\beta}(\vec{\beta}\cdot\vec{B})$$

- $\frac{\gamma^2}{\gamma+1}\vec{\beta}(\vec{\beta}\cdot\vec{B}) = 0$
- $\gamma\vec{B} = 0$

- $\vec{B}' = \frac{-\gamma}{c}(\vec{\beta}\times\vec{E})$

$$\vec{\beta}\times\vec{E} = -\beta\hat{z}\times\frac{\lambda}{2\pi\rho\varepsilon_0}\hat{\rho}$$

$$= -\frac{\beta\lambda}{2\pi\rho\varepsilon_0}(\hat{z}\times\hat{\rho})$$

$$\hat{\rho} = \cos\phi\,\hat{x} + \sin\phi\,\hat{y}$$

$$\hat{z}\times\hat{\rho} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & 1 \\ \cos\phi & \sin\phi & 0 \end{vmatrix}$$

$$= \hat{x}(0-\sin\phi) - \hat{y}(0-\cos\phi)$$

$$= -\sin\phi\,\hat{x} + \cos\phi\,\hat{y} = \hat{\phi}$$

$$\vec{B} = \frac{-\beta\lambda\gamma}{2\pi\rho\varepsilon_0 c}\hat{\phi}$$

d.) · General Lorentz transformations:

- $A_0' = \gamma\left(A_0 - \vec{\beta}\cdot\vec{A}\right)$, $(A_0, \underbrace{A_1, A_2, A_3}_{\vec{A}})$
- $p' = \gamma\left(p - \vec{\beta}\cdot\vec{J}\right)$

$p' = \gamma p$ $\quad \vec{J} = 0$ (In K-frame)

- $\boxed{p' = \frac{\gamma\lambda}{2\pi\varepsilon}\,\delta(p-\varepsilon)}$

- $J^0 = cp \quad (ct, \rho, \phi, z)$
- $J^1 = J^2 = J^3 = 0$
- $J^\alpha = (cp, \vec{J})$
- $\partial_\alpha F^{\alpha\beta} = \frac{4\pi}{c} J^\beta$ $\quad (+,-,-,-)$

$$\frac{4\pi}{c}J^0 = \partial_\alpha F^{\alpha 0} = \partial_1 F^{10} + \partial_2 F^{20} + \partial_3 F^{30} + \cancel{\partial_0 F^{00}}$$

$$= \partial_1\left(\frac{-E_x}{c}\right) + \partial_2\left(\frac{-E_y}{c}\right) + \partial_3\left(\frac{-E_z}{c}\right)$$

- $-4\pi\rho = \nabla\cdot E$

Ampere's Law:

Gauss's Law: $\vec{\nabla}\cdot\vec{E} = 4\pi\rho$

$$J^{1'} = -\gamma c \rho \beta$$

$$J^{2'} = 0$$

$$J^{3'} = 0$$

$$J^{1'} = \frac{-\gamma^2 c \beta \lambda}{2\pi\varepsilon}\,\delta(\rho-\varepsilon)$$

$$\vec{J} = \frac{-\gamma^2 c \beta \lambda}{2\pi\varepsilon}\,\delta(\rho-\varepsilon)\,\hat{z}$$

- $\lambda' = \int \vec{\rho}' \cdot d\vec{A}$

- $\lambda' = \frac{\gamma\lambda}{2\pi\varepsilon} \int_0^{2\pi}\int_0^{\infty} \rho'\,\delta(\rho-\varepsilon)\,d\rho' d\phi'$

$$\lambda' = \gamma\lambda$$

- $I' = \int \vec{J} \cdot d\vec{A}$

$$I' = \frac{-\gamma^2 c \beta \lambda}{2\pi\varepsilon} \int_0^{2\pi}\int_0^{\infty} \rho'\,\delta(\rho-\varepsilon)\,d\rho' d\phi'$$

$$I' = -\gamma^2 c \beta \lambda$$

The results make sense because moving frames cause things to move in opposite directions.